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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 12.14

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} (\sqrt{x^2 + ax} - \sqrt{x^2 + bx}) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{(\sqrt{x^2 + ax} - \sqrt{x^2 + bx})(\sqrt{x^2 + ax} + \sqrt{x^2 + bx})}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{(\sqrt{x^2 + ax})^2 - (\sqrt{x^2 + bx})^2}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{(x^2 + ax) - (x^2 + bx)}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{ax - bx}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \right)
 \end{aligned}$$

Limiet voor $x \rightarrow +\infty$

$$\begin{aligned}
 &= \lim_{x \rightarrow +\infty} \left(\frac{x(a - b)}{|x|\sqrt{1 + \frac{a}{x}} + |x|\sqrt{1 + \frac{b}{x}}} \right) \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{x(a - b)}{x\sqrt{1 + \frac{a}{x}} + x\sqrt{1 + \frac{b}{x}}} \right) \\
 &= \lim_{x \rightarrow +\infty} \left(\frac{a - b}{\sqrt{1 + \frac{a}{x}} + \sqrt{1 + \frac{b}{x}}} \right) \\
 &= \frac{a - b}{\sqrt{1 + \frac{a}{\infty}} + \sqrt{1 + \frac{b}{\infty}}} \\
 &= \frac{a - b}{\sqrt{1 + 0} + \sqrt{1 + 0}} \\
 &= \frac{a - b}{2\sqrt{1}} \\
 &= \frac{a - b}{2}
 \end{aligned}$$

Limiet voor $x \rightarrow -\infty$

$$= \lim_{x \rightarrow -\infty} \left(\frac{x(a - b)}{|x|\sqrt{1 + \frac{a}{x}} + |x|\sqrt{1 + \frac{b}{x}}} \right)$$

$$\begin{aligned}
&= \lim_{x \rightarrow -\infty} \left(\frac{x(a-b)}{-x\sqrt{1+\frac{a}{x}} - x\sqrt{1+\frac{b}{x}}} \right) \\
&= \lim_{x \rightarrow -\infty} \left(\frac{a-b}{-\left(\sqrt{1+\frac{a}{x}} + \sqrt{1+\frac{b}{x}}\right)} \right) \\
&= \frac{a-b}{-\left(\sqrt{1+\frac{a}{\infty}} + \sqrt{1+\frac{b}{\infty}}\right)} \\
&= \frac{a-b}{-(\sqrt{1+0} + \sqrt{1+0})} \\
&= \frac{a-b}{-2\sqrt{1}} \\
&= \frac{a-b}{-2} \\
&= \frac{b-a}{2}
\end{aligned}$$

References